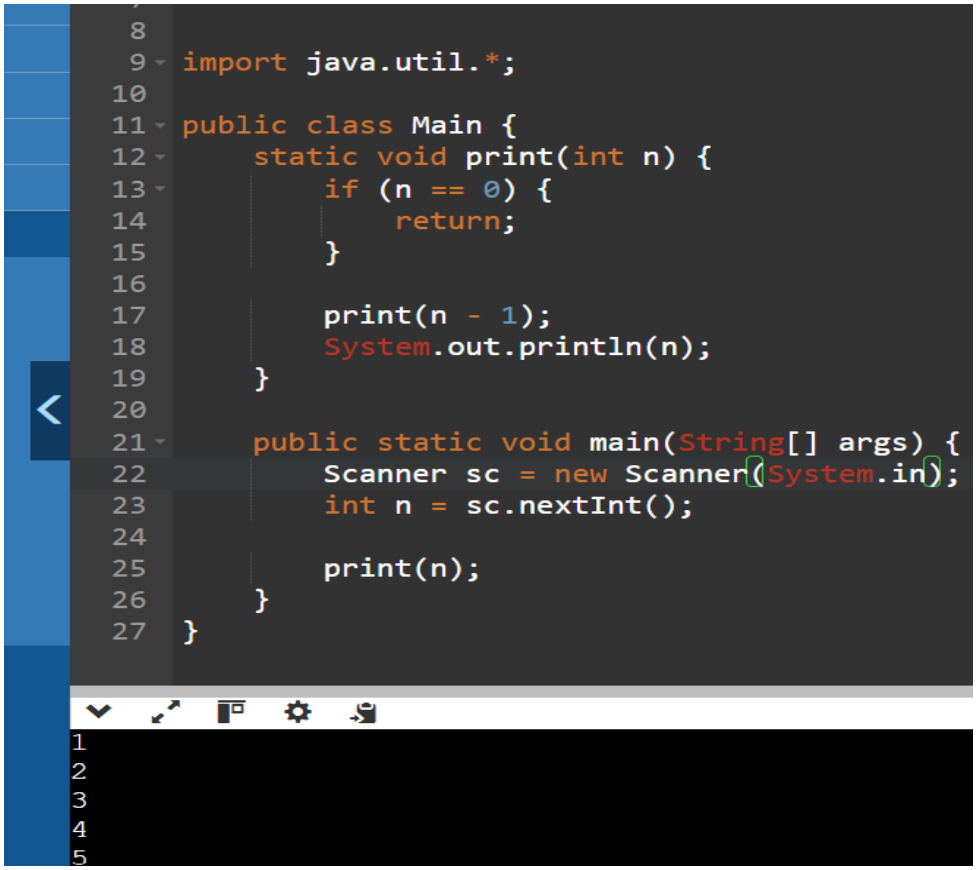
ASSIGNMENT

1)Print Numbers From 1 to N



```java

import java.util.*;

public class Main {
    static void print(int n) {
        if (n == 0) {
            return;
        }

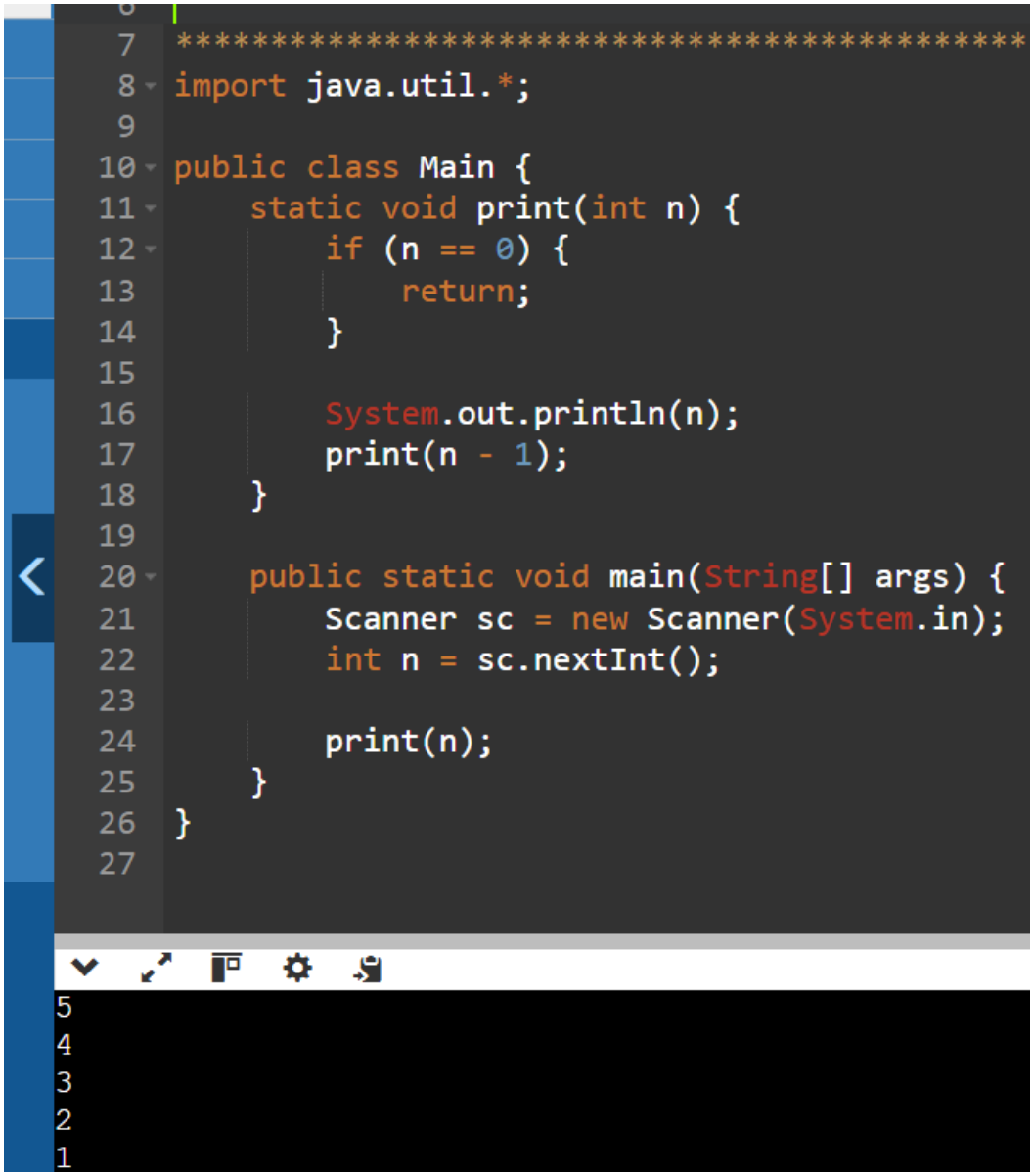
        print(n - 1);
        System.out.println(n);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        print(n);
    }
}
```

```
1
2
3
4
5
```

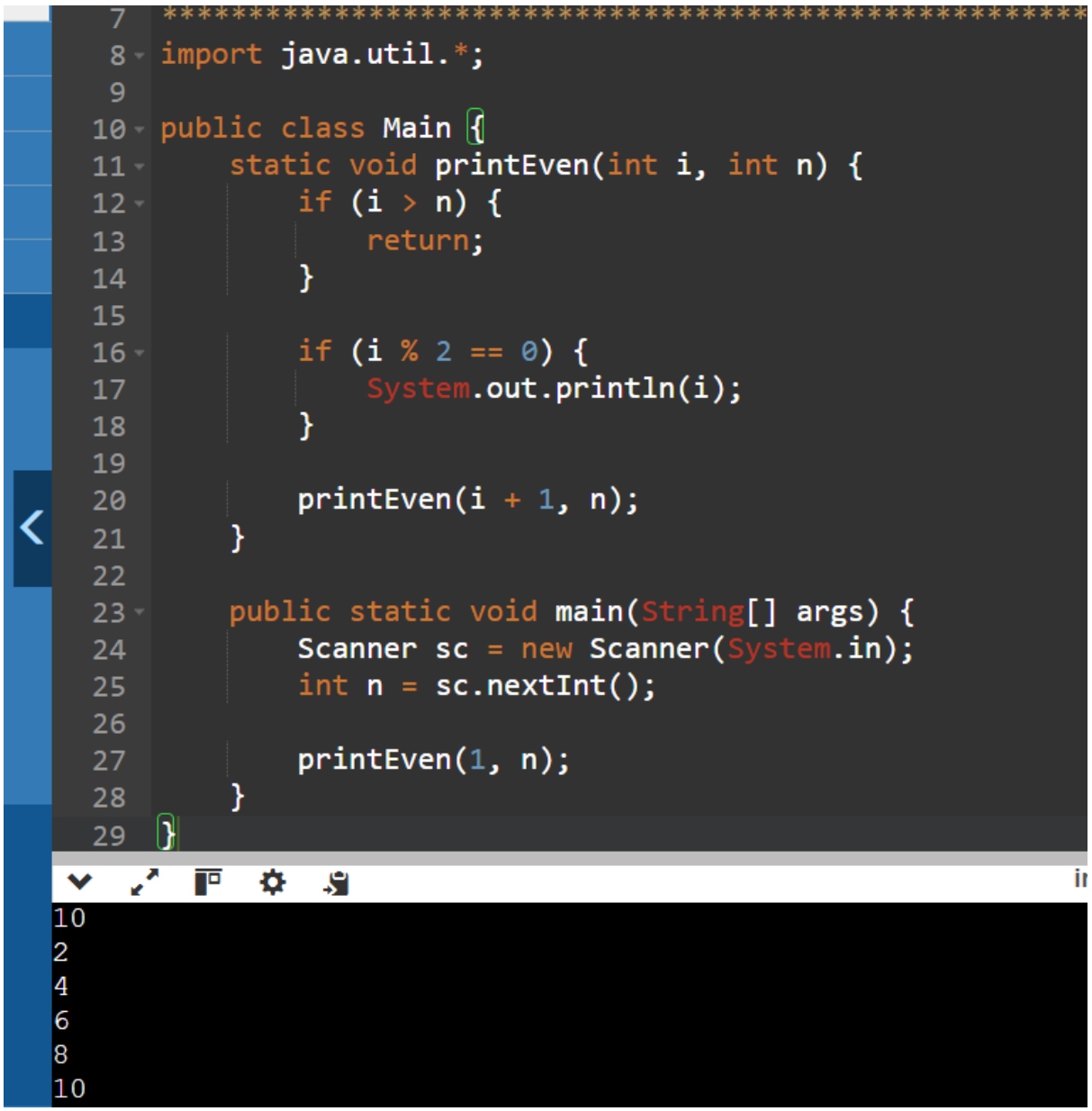
2)Print Numbers from N to 1



```
7  ********************************************
8  import java.util.*;
9
10 public class Main {
11     static void print(int n) {
12         if (n == 0) {
13             return;
14         }
15
16         System.out.println(n);
17         print(n - 1);
18     }
19
20     public static void main(String[] args) {
21         Scanner sc = new Scanner(System.in);
22         int n = sc.nextInt();
23
24         print(n);
25     }
26 }
27
```

```
5
4
3
2
1
```

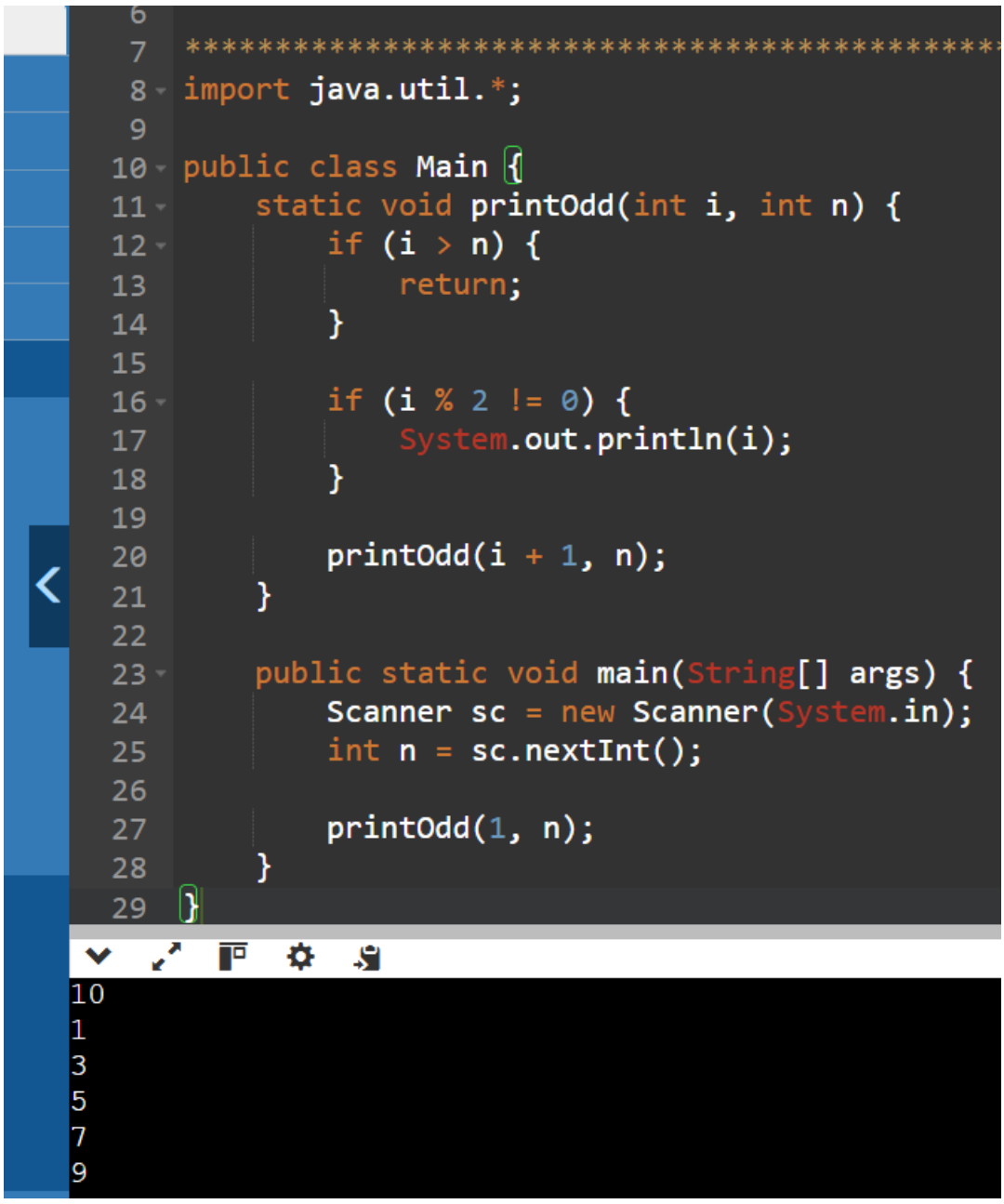
3)Print even Numbers from 1 to N



```
7  ************************************************
8  import java.util.*;
9
10 public class Main {
11     static void printEven(int i, int n) {
12         if (i > n) {
13             return;
14         }
15
16         if (i % 2 == 0) {
17             System.out.println(i);
18         }
19
20         printEven(i + 1, n);
21     }
22
23     public static void main(String[] args) {
24         Scanner sc = new Scanner(System.in);
25         int n = sc.nextInt();
26
27         printEven(1, n);
28     }
29 }
```

```
10
2
4
6
8
10
```

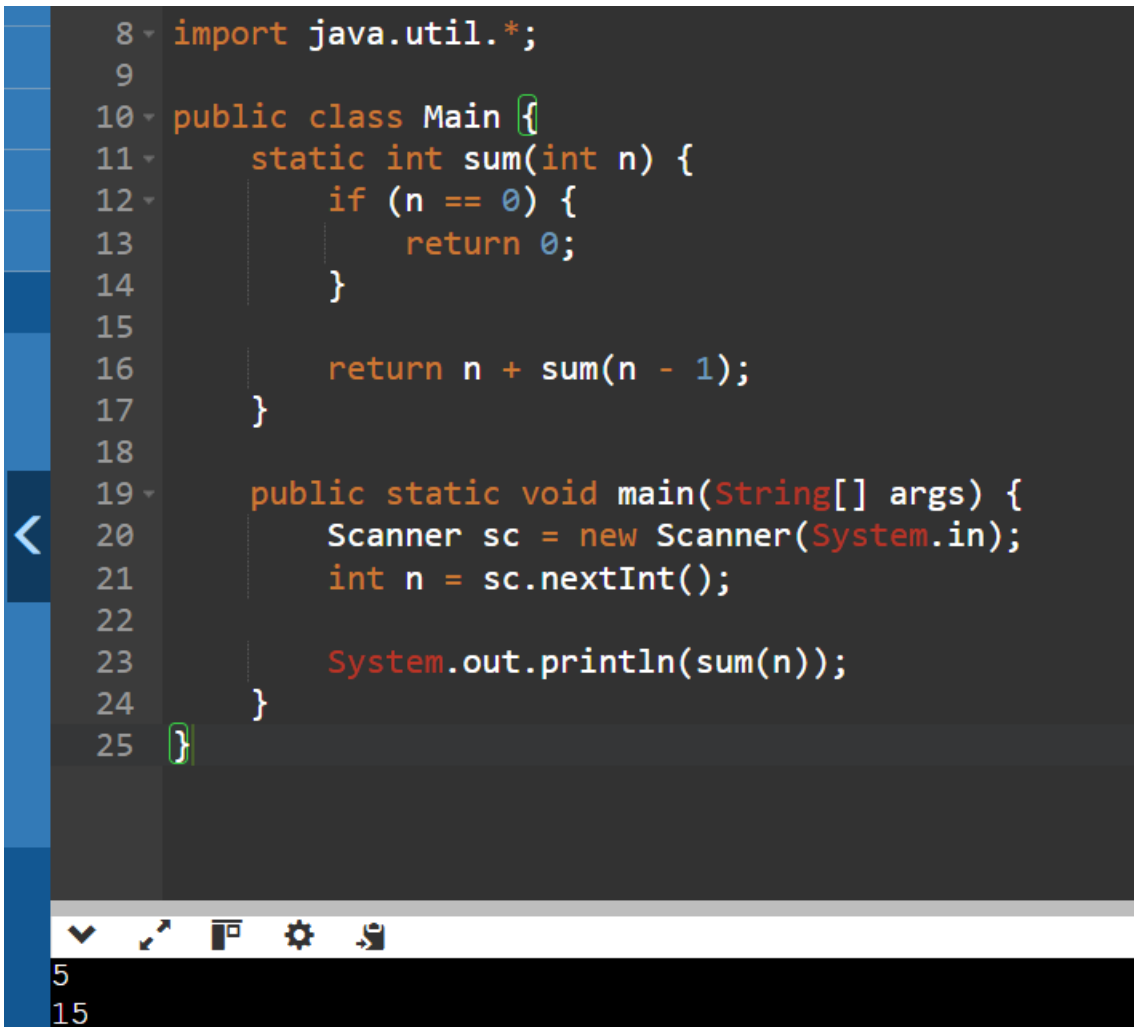
4)Print Odd Numbers from 1 to N



```
6
7  ********************************************
8  import java.util.*;
9
10 public class Main {
11     static void printOdd(int i, int n) {
12         if (i > n) {
13             return;
14         }
15
16         if (i % 2 != 0) {
17             System.out.println(i);
18         }
19
20         printOdd(i + 1, n);
21     }
22
23     public static void main(String[] args) {
24         Scanner sc = new Scanner(System.in);
25         int n = sc.nextInt();
26
27         printOdd(1, n);
28     }
29 }
```

```
10
1
3
5
7
9
```

5)Sum of First N Natural Numbers



```java
import java.util.*;

public class Main {
    static int sum(int n) {
        if (n == 0) {
            return 0;
        }

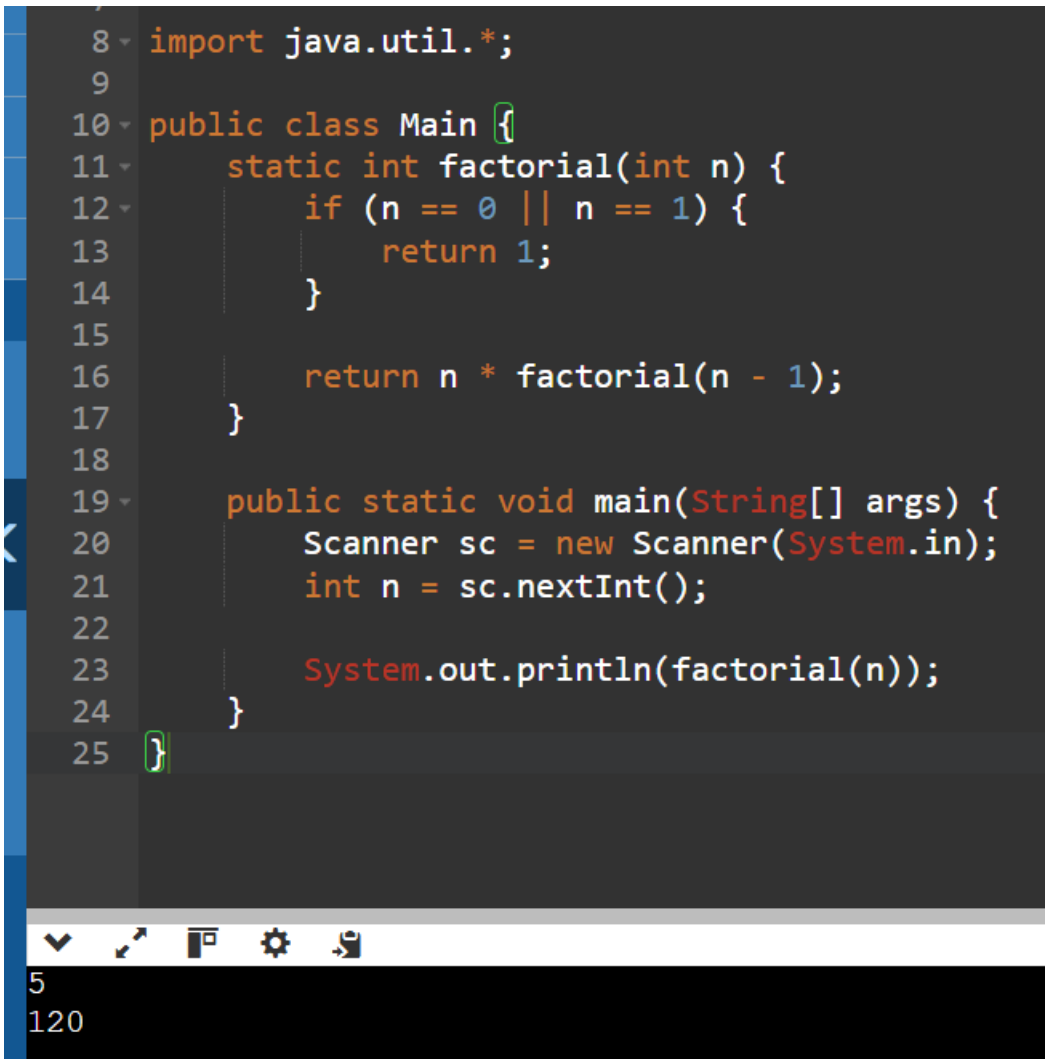
        return n + sum(n - 1);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        System.out.println(sum(n));
    }
}
```

```
5
15
```

6)Factorial of a Number



```java
import java.util.*;

public class Main {
    static int factorial(int n) {
        if (n == 0 || n == 1) {
            return 1;
        }

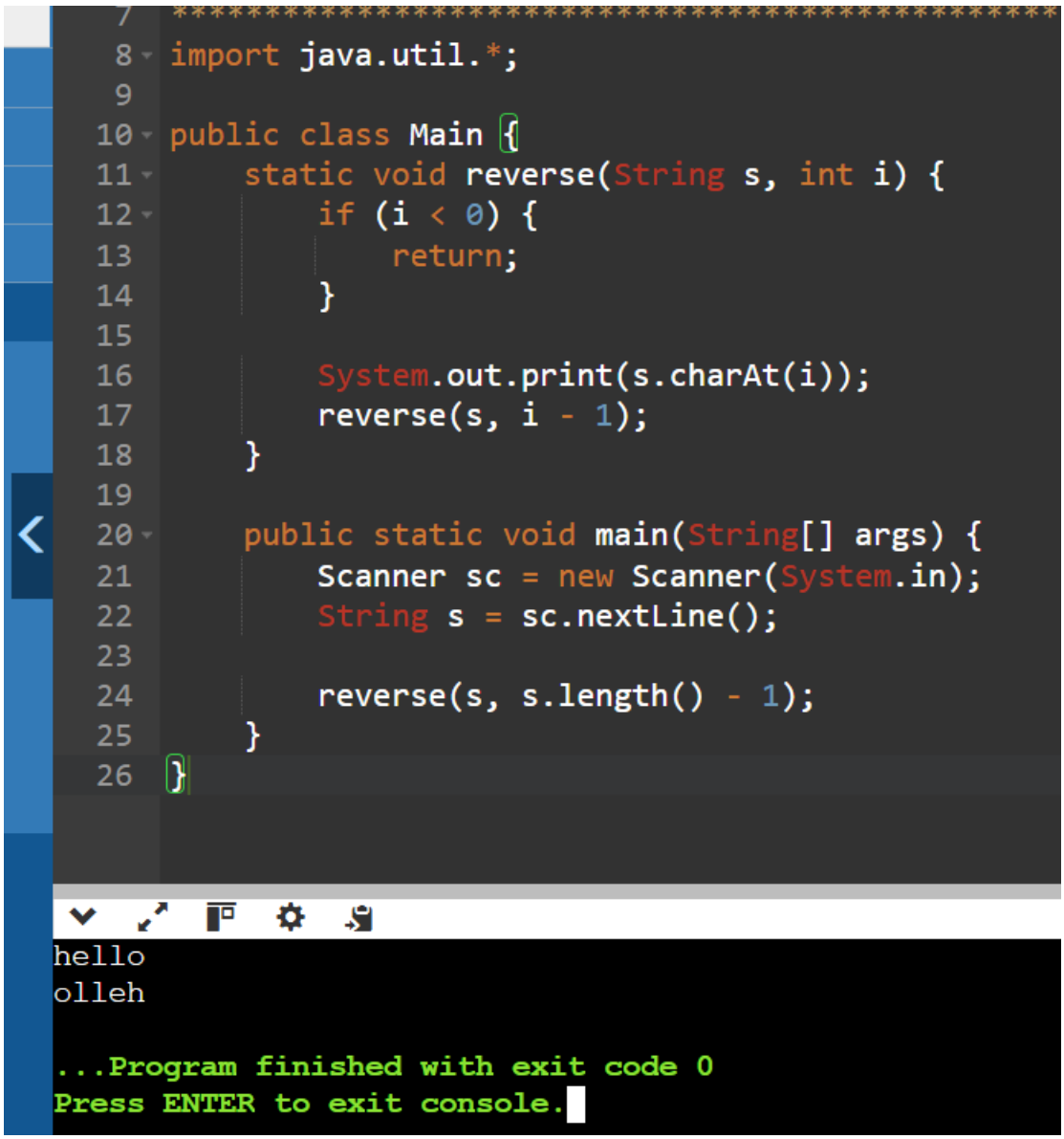
        return n * factorial(n - 1);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        System.out.println(factorial(n));
    }
}
```

```
5
120
```

7)Reverse a String



```
import java.util.*;

public class Main {
    static void reverse(String s, int i) {
        if (i < 0) {
            return;
        }

        System.out.print(s.charAt(i));
        reverse(s, i - 1);
    }

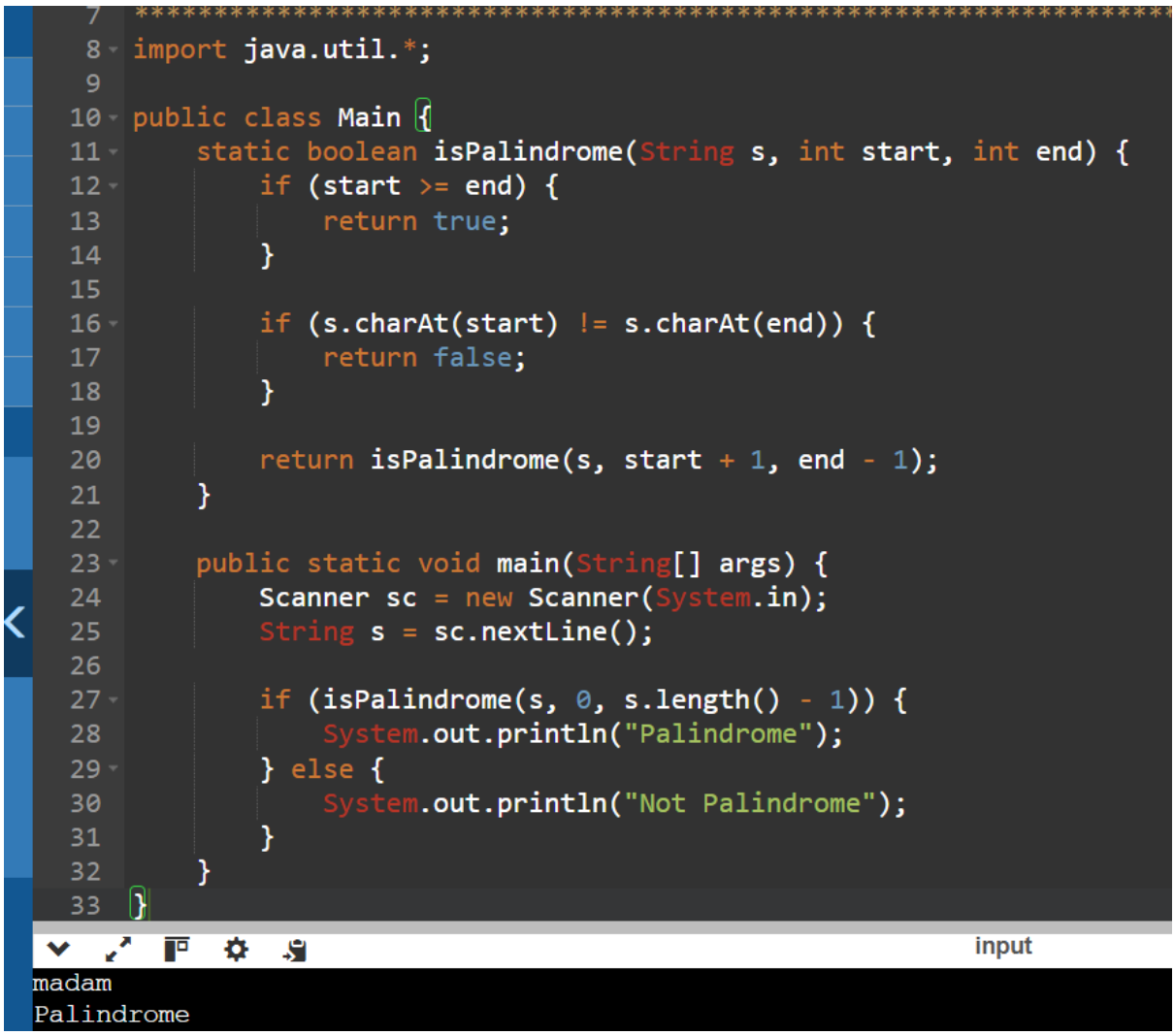
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();

        reverse(s, s.length() - 1);
    }
}
```

```
hello
olleh

...Program finished with exit code 0
Press ENTER to exit console.
```

8)Check whether a string is in Pallindrome



```
import java.util.*;

public class Main {
    static boolean isPalindrome(String s, int start, int end) {
        if (start >= end) {
            return true;
        }

        if (s.charAt(start) != s.charAt(end)) {
            return false;
        }

        return isPalindrome(s, start + 1, end - 1);
    }

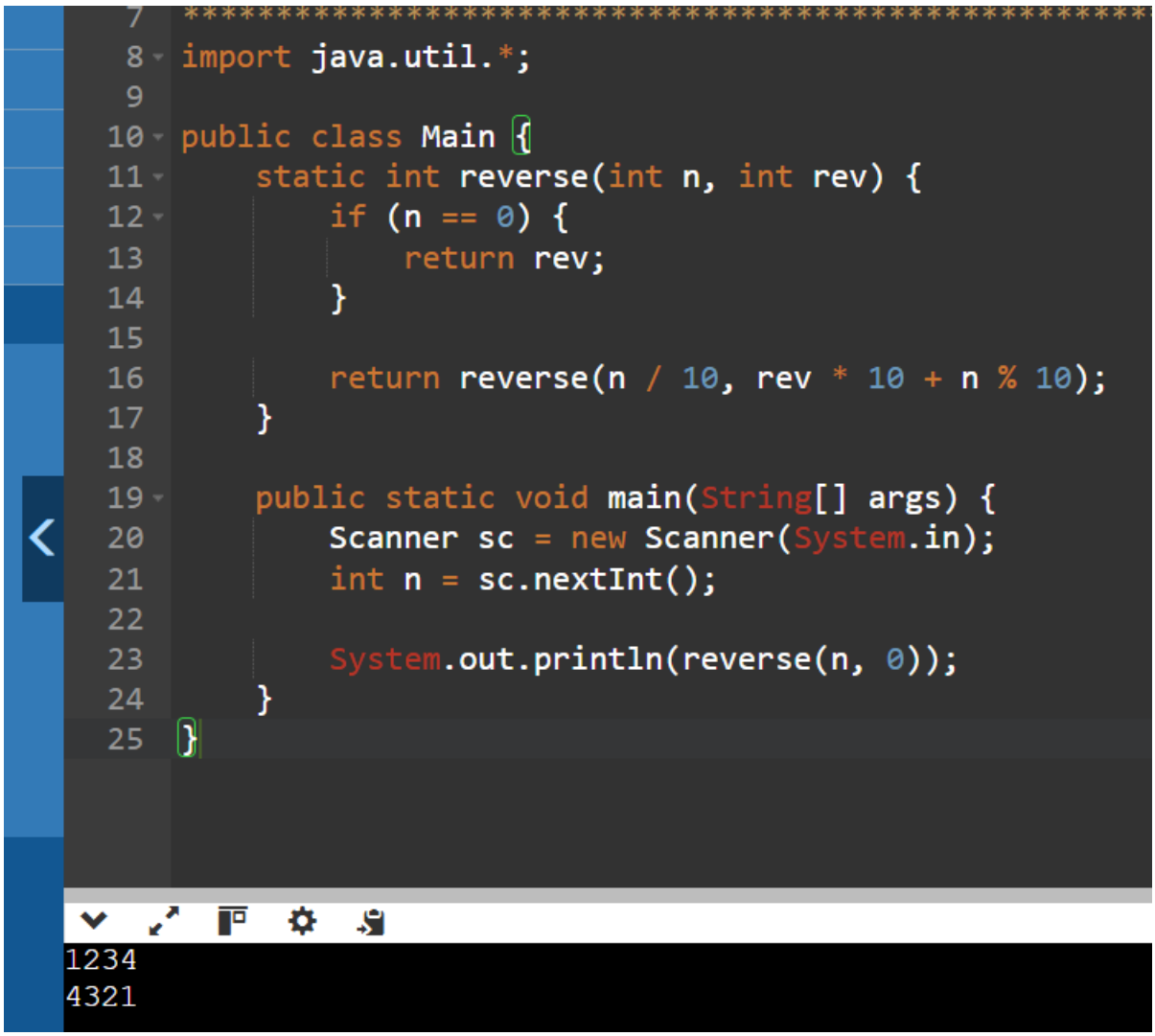
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();

        if (isPalindrome(s, 0, s.length() - 1)) {
            System.out.println("Palindrome");
        } else {
            System.out.println("Not Palindrome");
        }
    }
}
```

input

```
madam
Palindrome
```

9)Reverse a Number



```
**********************************************************
import java.util.*;

public class Main {
    static int reverse(int n, int rev) {
        if (n == 0) {
            return rev;
        }

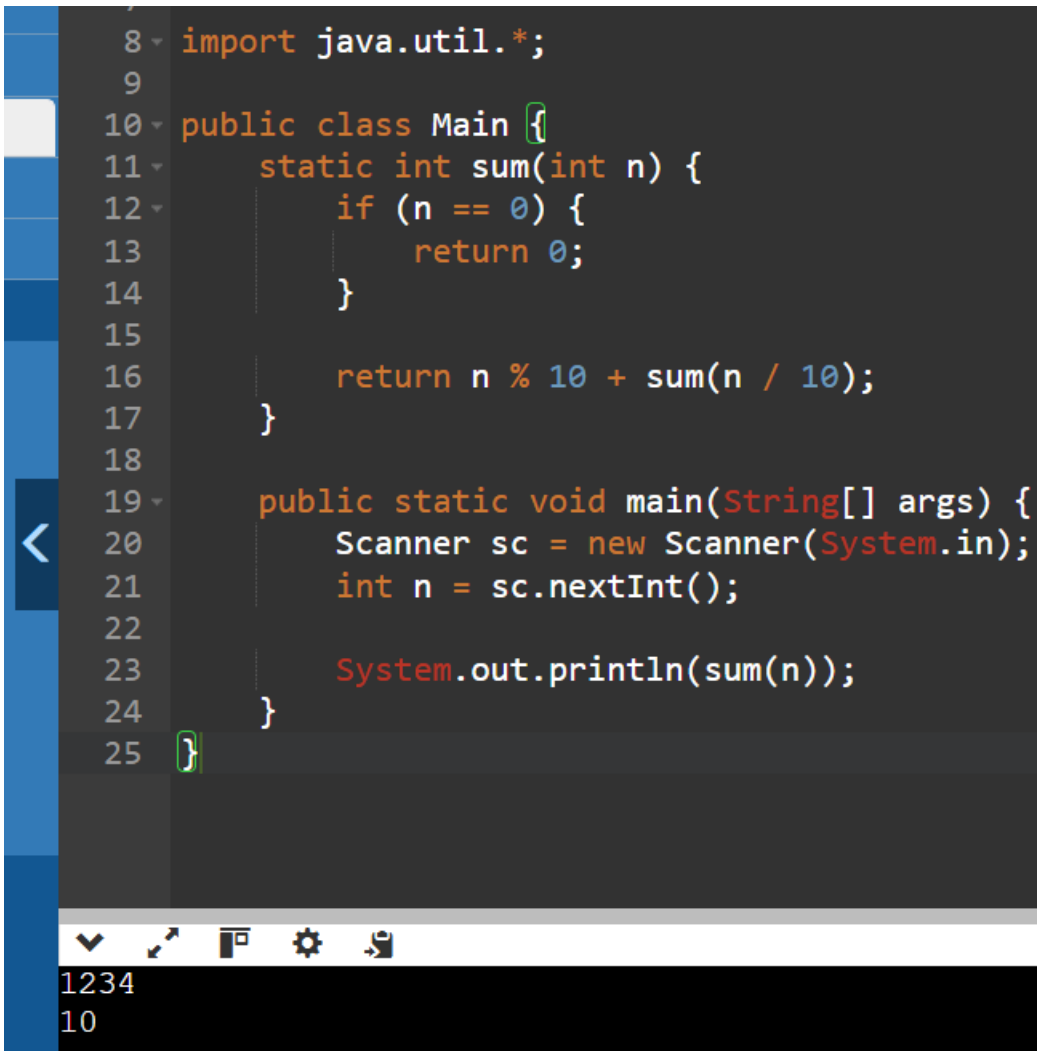
        return reverse(n / 10, rev * 10 + n % 10);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        System.out.println(reverse(n, 0));
    }
}
```

```
1234
4321
```

10)Sum of Digits



```java
import java.util.*;

public class Main {
    static int sum(int n) {
        if (n == 0) {
            return 0;
        }

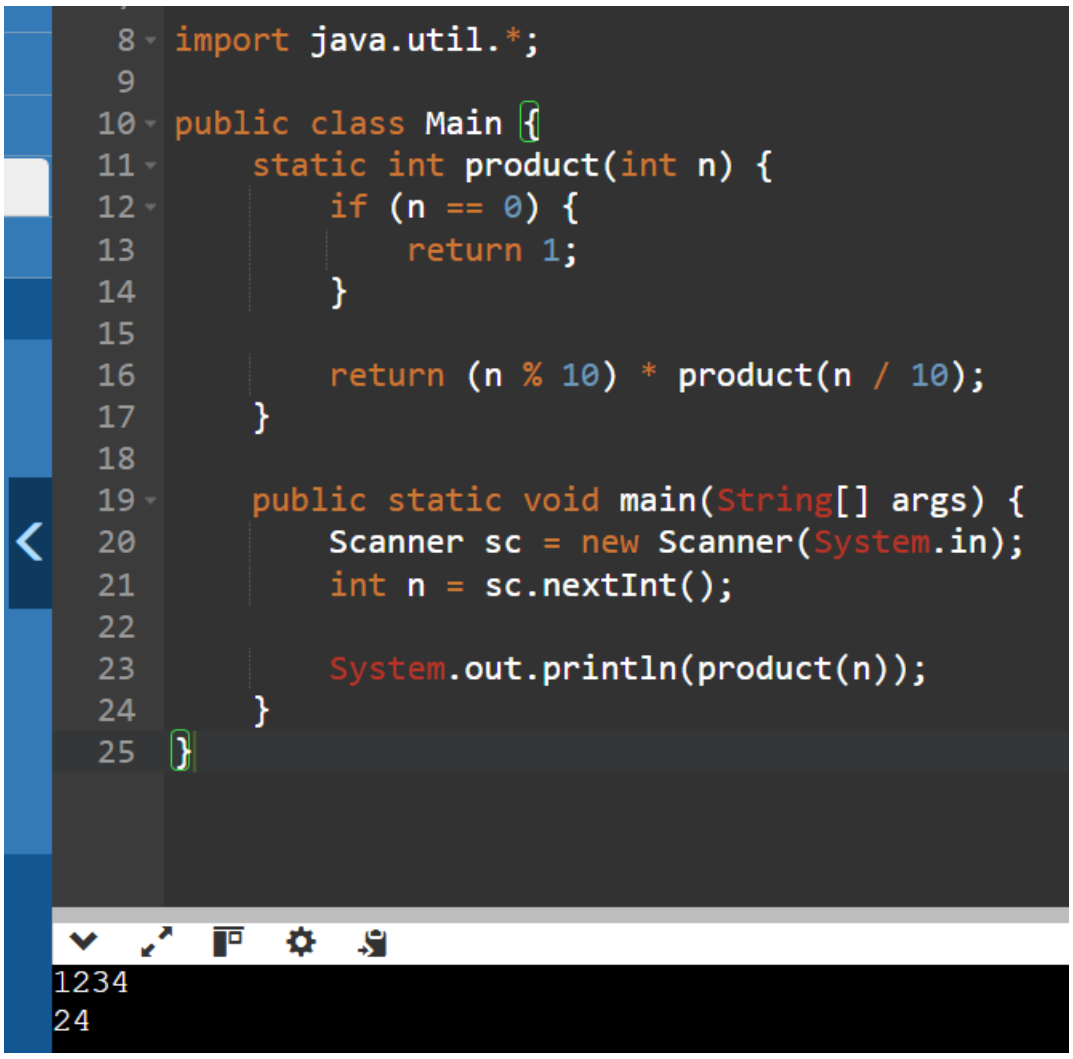
        return n % 10 + sum(n / 10);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        System.out.println(sum(n));
    }
}
```

```
1234
10
```

11)Product of Digits



```java
import java.util.*;

public class Main {
    static int product(int n) {
        if (n == 0) {
            return 1;
        }

        return (n % 10) * product(n / 10);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        System.out.println(product(n));
    }
}
```

```
1234
24
```

12)Count Digits of a Number



```java
import java.util.*;

public class Main {
    static int count(int n) {
        if (n == 0) {
            return 0;
        }

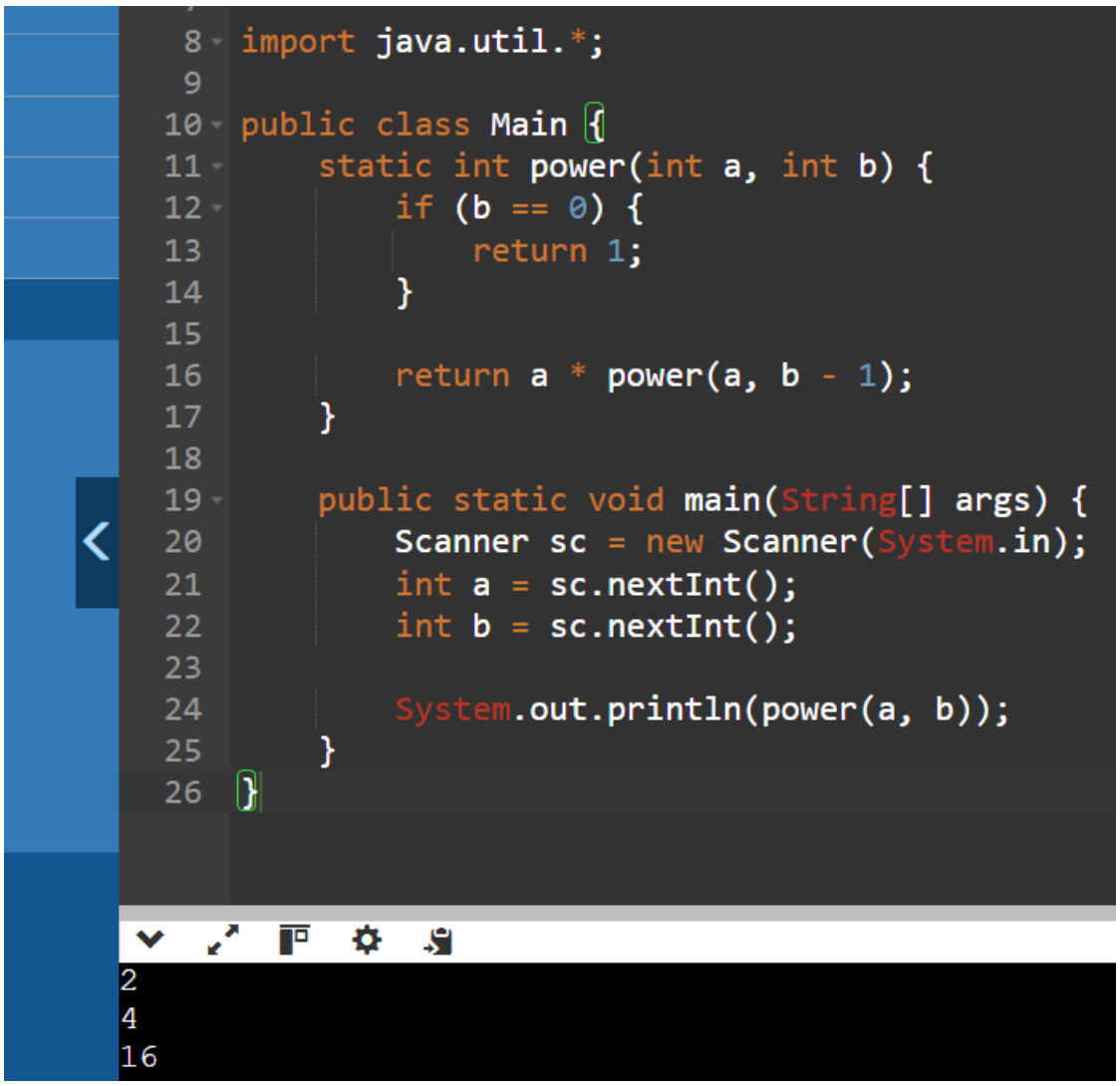
        return 1 + count(n / 10);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        if (n == 0) {
            System.out.println(1);
        } else {
            System.out.println(count(Math.abs(n)));
        }
    }
}
```

```
12345
5
```

13)Power of a Number



```java
import java.util.*;

public class Main {
    static int power(int a, int b) {
        if (b == 0) {
            return 1;
        }

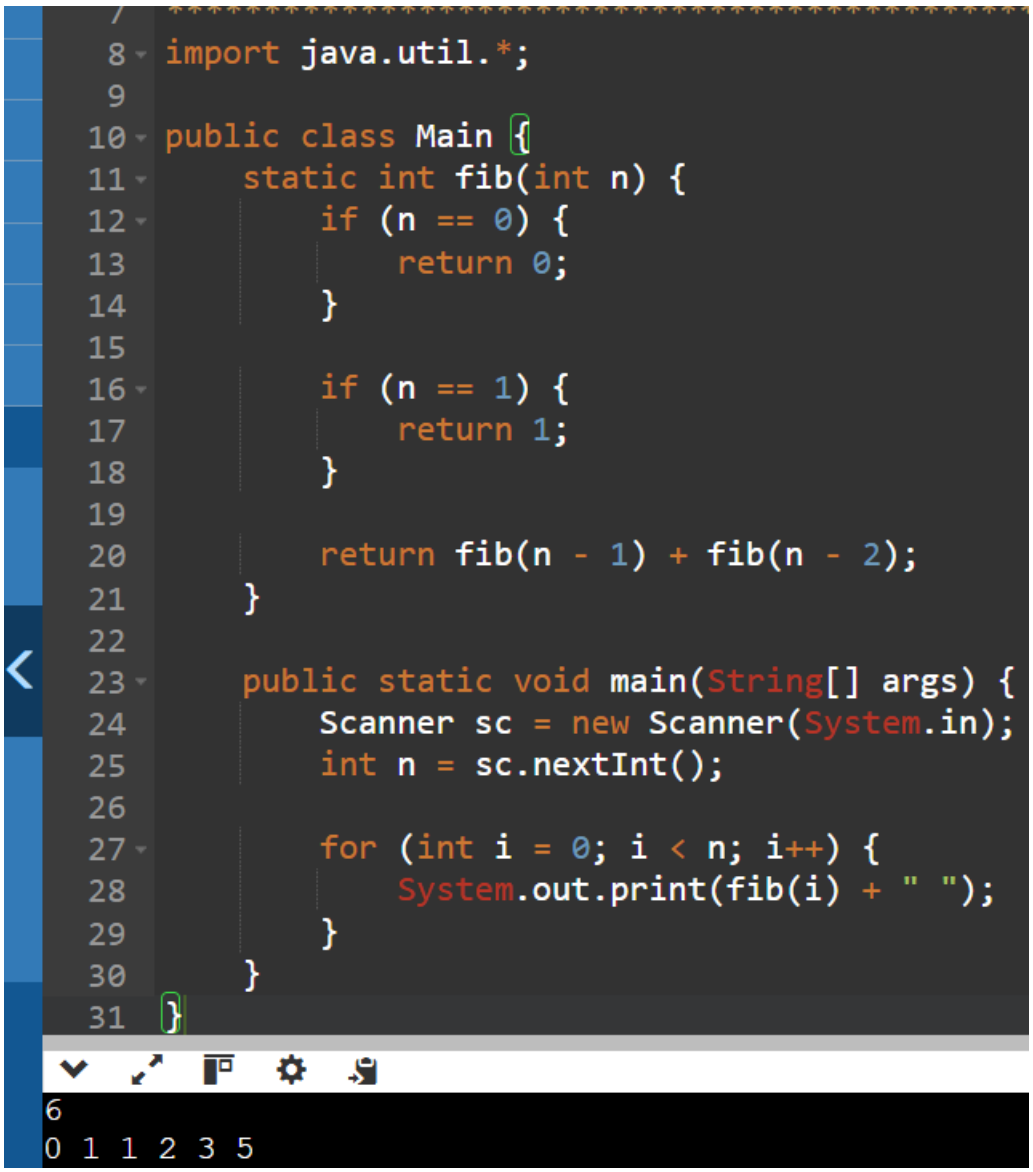
        return a * power(a, b - 1);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int a = sc.nextInt();
        int b = sc.nextInt();

        System.out.println(power(a, b));
    }
}
```

```
2
4
16
```

14)Fibonacci Series



```java
import java.util.*;

public class Main {
    static int fib(int n) {
        if (n == 0) {
            return 0;
        }

        if (n == 1) {
            return 1;
        }

        return fib(n - 1) + fib(n - 2);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        for (int i = 0; i < n; i++) {
            System.out.print(fib(i) + " ");
        }
    }
}
```

```
6
0 1 1 2 3 5
```

15)GCD of two Numbers

```java
import java.util.*;

public class Main {
    static int gcd(int a, int b) {
        if (b == 0) {
            return a;
        }

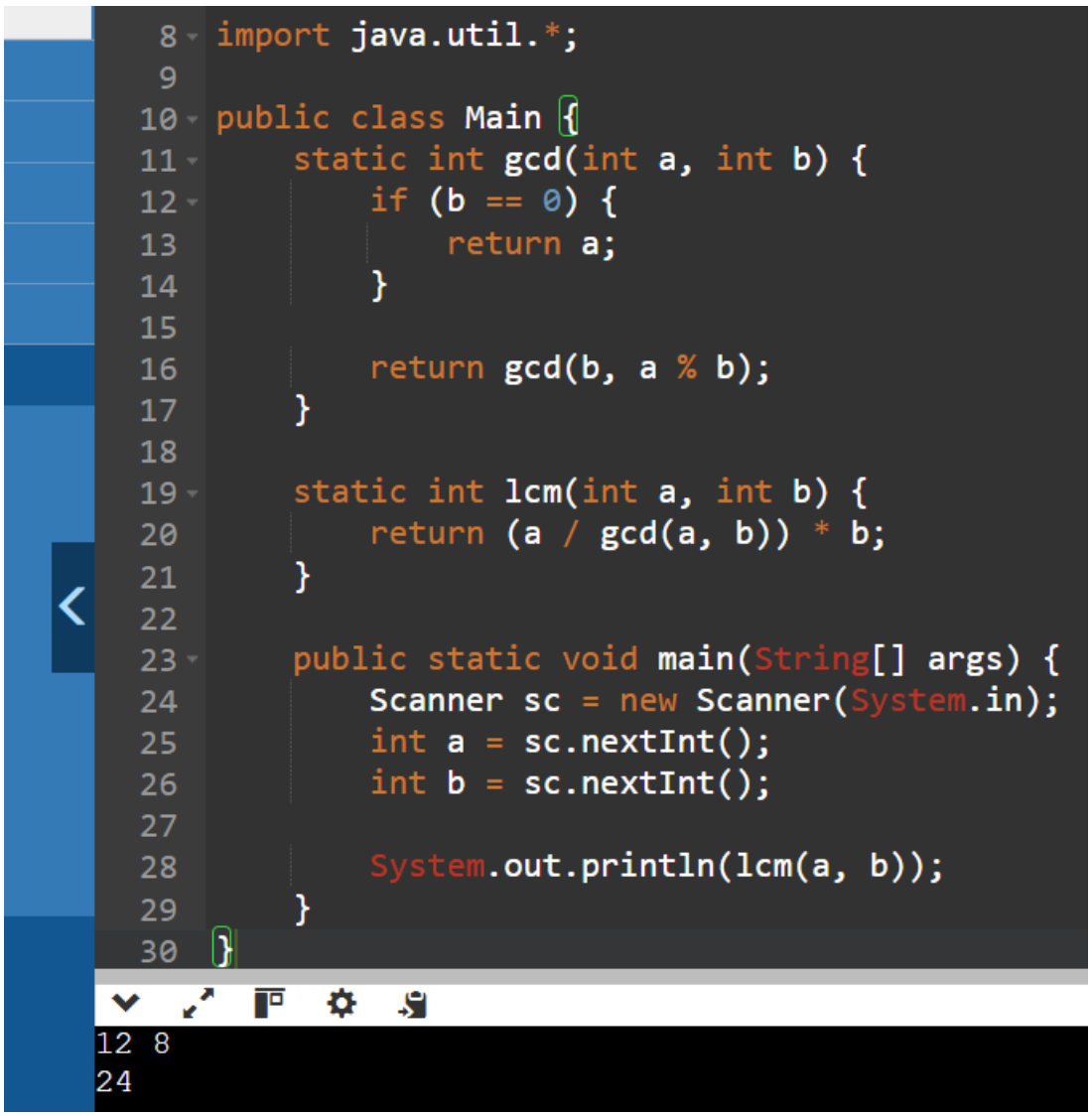
        return gcd(b, a % b);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int a = sc.nextInt();
        int b = sc.nextInt();

        System.out.println(gcd(a, b));
    }
}
```

```
12 8
4
```

16)LCM of two Numbers



```java
import java.util.*;

public class Main {
    static int gcd(int a, int b) {
        if (b == 0) {
            return a;
        }

        return gcd(b, a % b);
    }

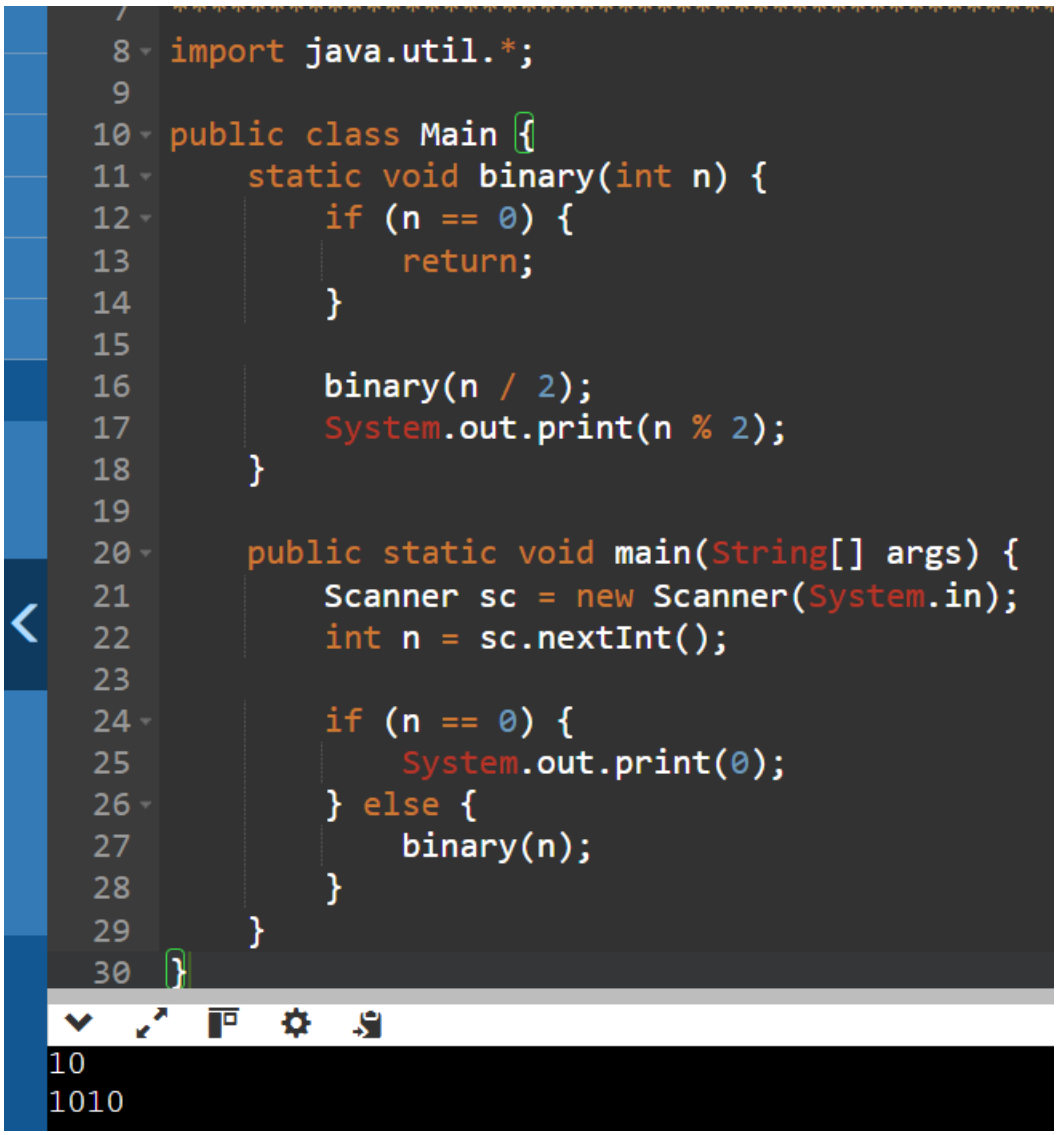
    static int lcm(int a, int b) {
        return (a / gcd(a, b)) * b;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int a = sc.nextInt();
        int b = sc.nextInt();

        System.out.println(lcm(a, b));
    }
}
```

```
12 8
24
```

17)Decimal to Binary



```java
import java.util.*;

public class Main {
    static void binary(int n) {
        if (n == 0) {
            return;
        }

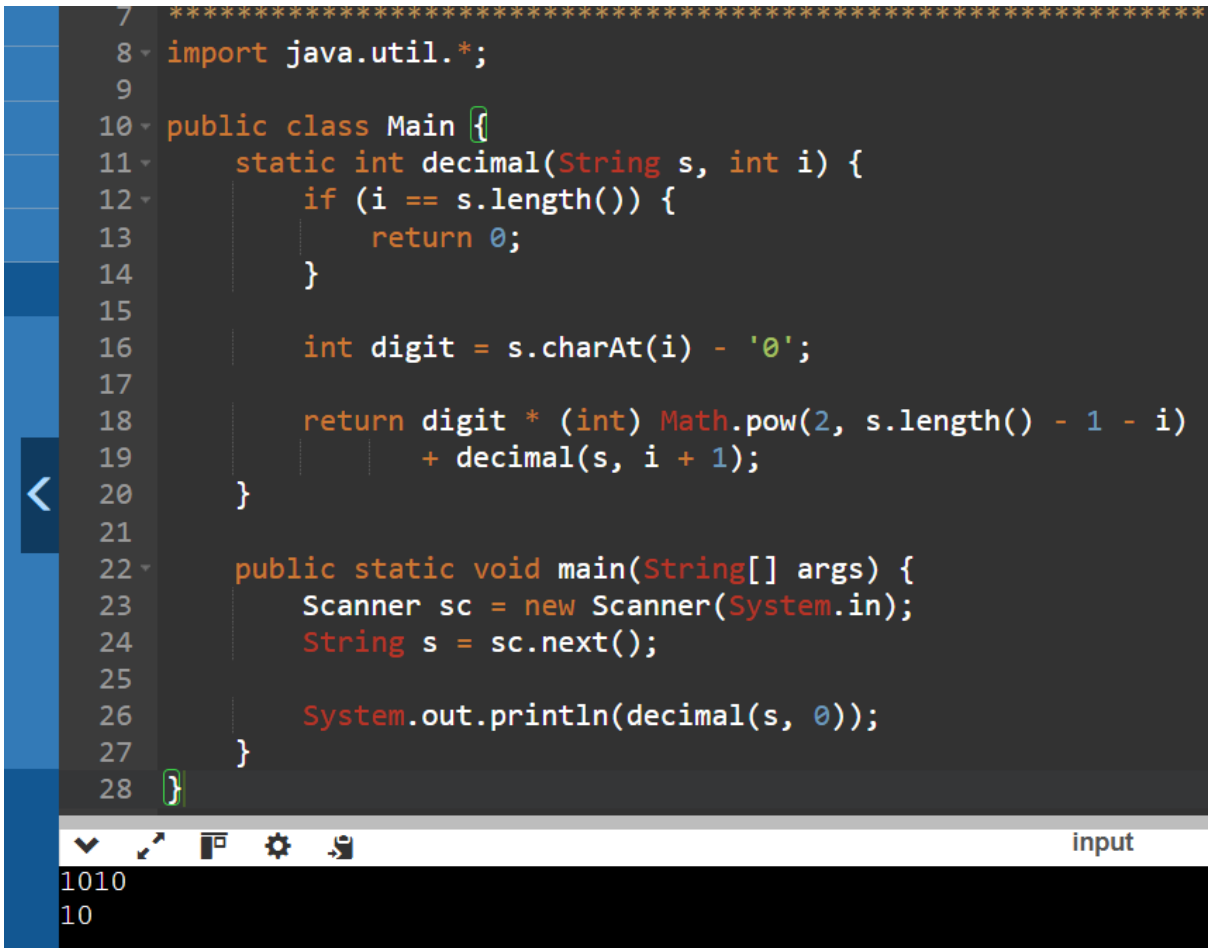
        binary(n / 2);
        System.out.print(n % 2);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        if (n == 0) {
            System.out.print(0);
        } else {
            binary(n);
        }
    }
}
```

```
10
1010
```

18)Binary to Decimal



```
import java.util.*;

public class Main {
    static int decimal(String s, int i) {
        if (i == s.length()) {
            return 0;
        }

        int digit = s.charAt(i) - '0';

        return digit * (int) Math.pow(2, s.length() - 1 - i)
                + decimal(s, i + 1);
    }

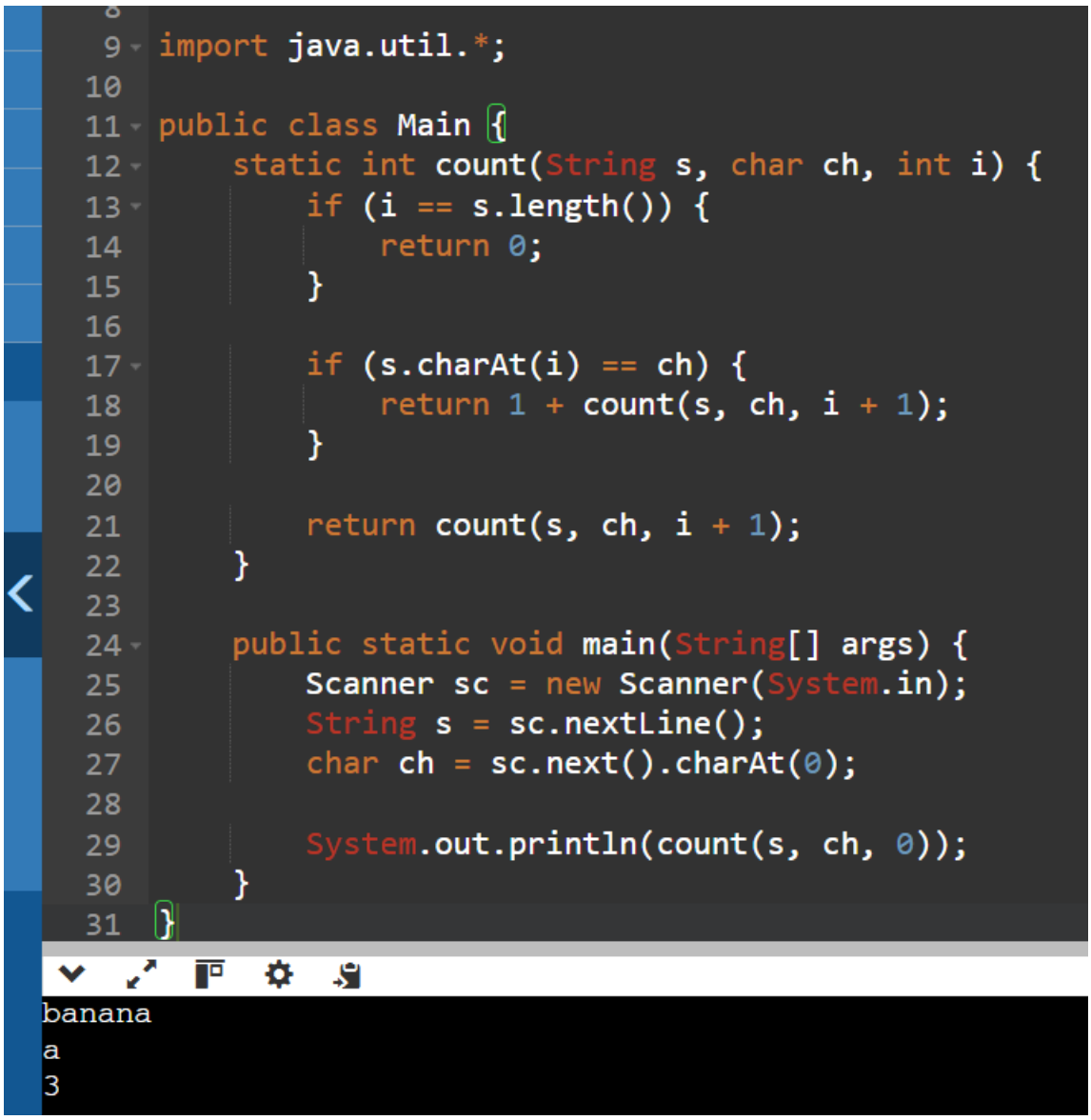
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s = sc.next();

        System.out.println(decimal(s, 0));
    }
}
```

input

```
1010
10
```

19) Count Occurrence of a character in a String



```java
import java.util.*;

public class Main {
    static int count(String s, char ch, int i) {
        if (i == s.length()) {
            return 0;
        }

        if (s.charAt(i) == ch) {
            return 1 + count(s, ch, i + 1);
        }

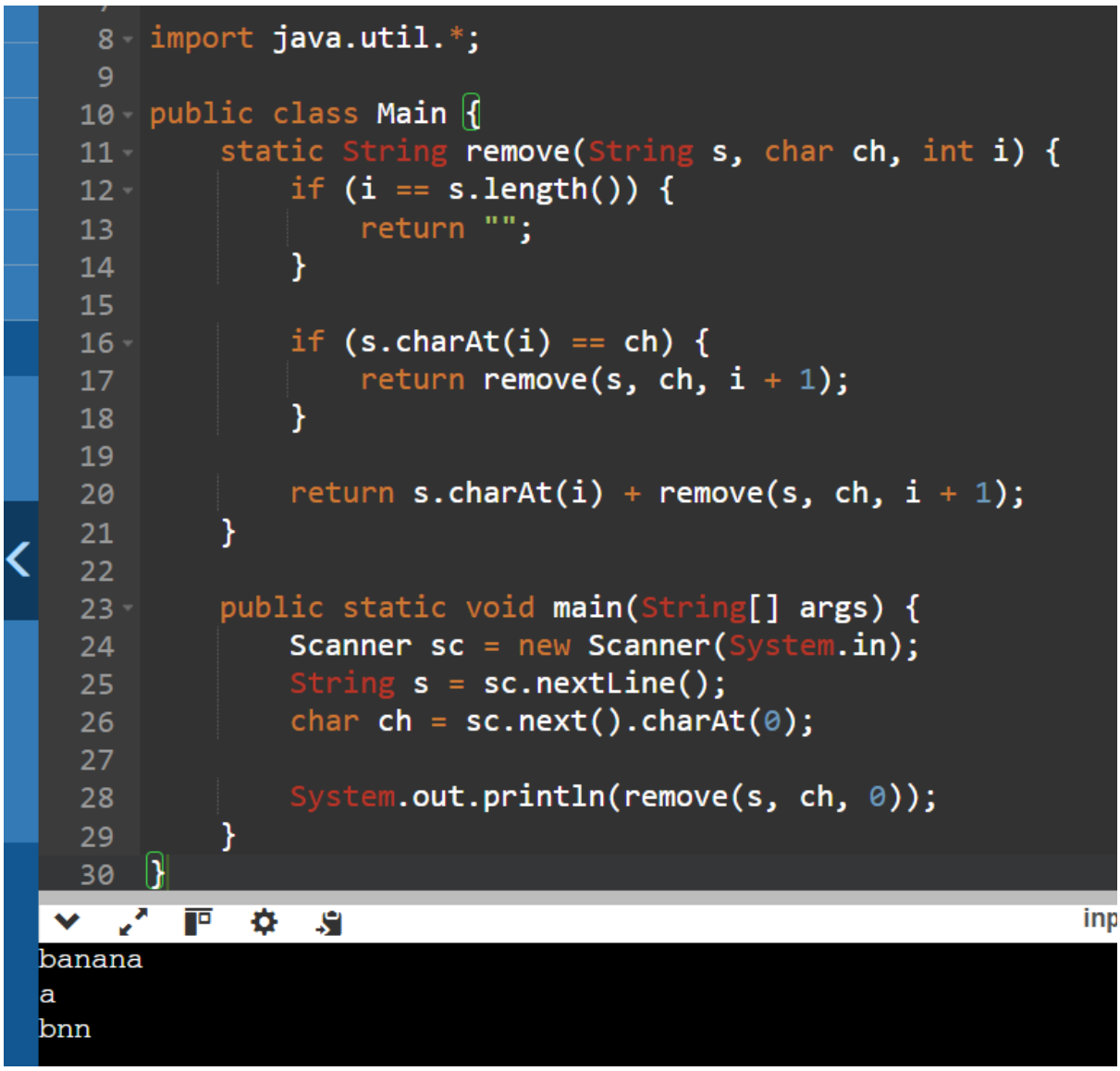
        return count(s, ch, i + 1);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        char ch = sc.next().charAt(0);

        System.out.println(count(s, ch, 0));
    }
}
```

```
banana
a
3
```

20) Remove a Character from a String



```java
import java.util.*;

public class Main {
    static String remove(String s, char ch, int i) {
        if (i == s.length()) {
            return "";
        }

        if (s.charAt(i) == ch) {
            return remove(s, ch, i + 1);
        }

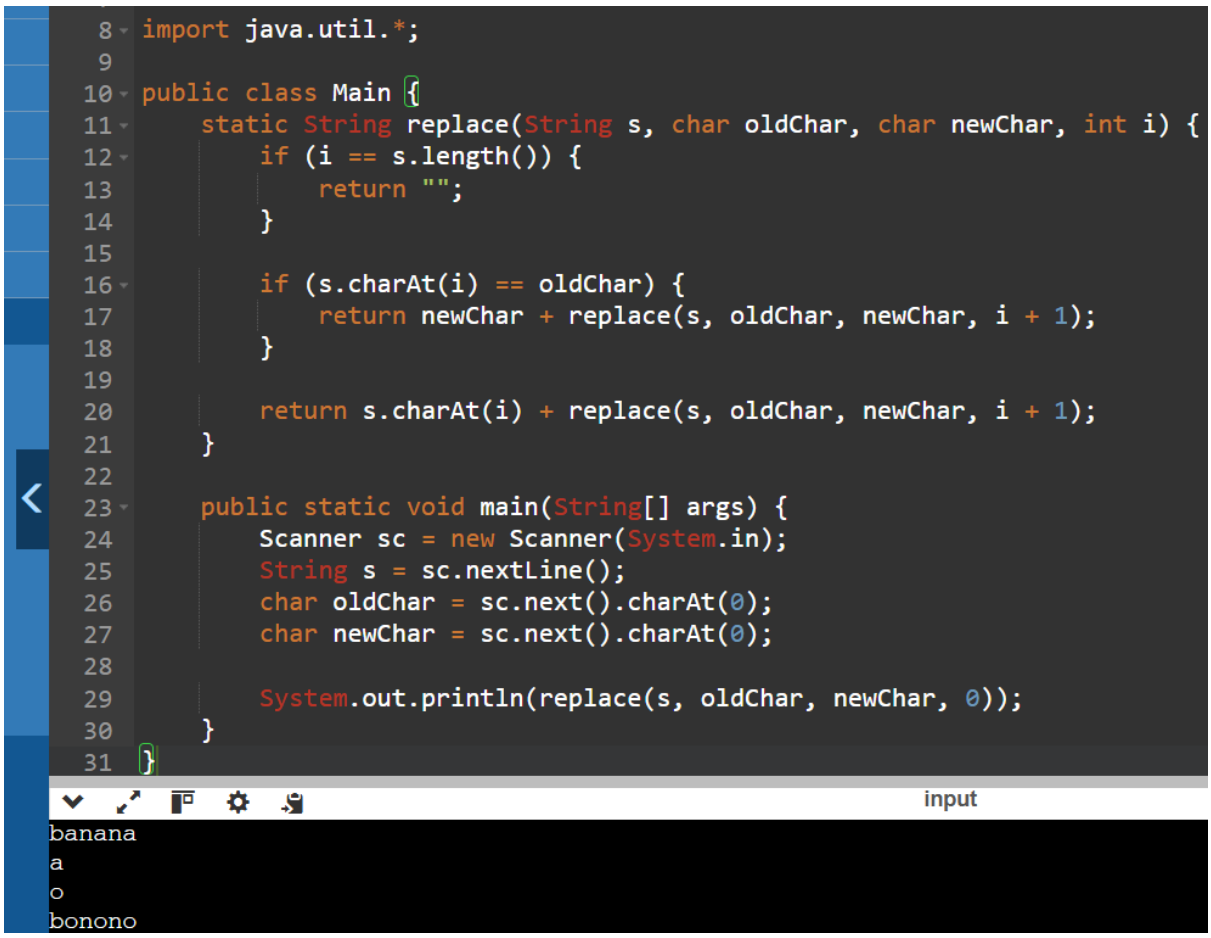
        return s.charAt(i) + remove(s, ch, i + 1);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        char ch = sc.next().charAt(0);

        System.out.println(remove(s, ch, 0));
    }
}
```

```
banana
a
bnn
```

21) Replace a Character in a String



```java
import java.util.*;

public class Main {
    static String replace(String s, char oldChar, char newChar, int i) {
        if (i == s.length()) {
            return "";
        }

        if (s.charAt(i) == oldChar) {
            return newChar + replace(s, oldChar, newChar, i + 1);
        }

        return s.charAt(i) + replace(s, oldChar, newChar, i + 1);
    }

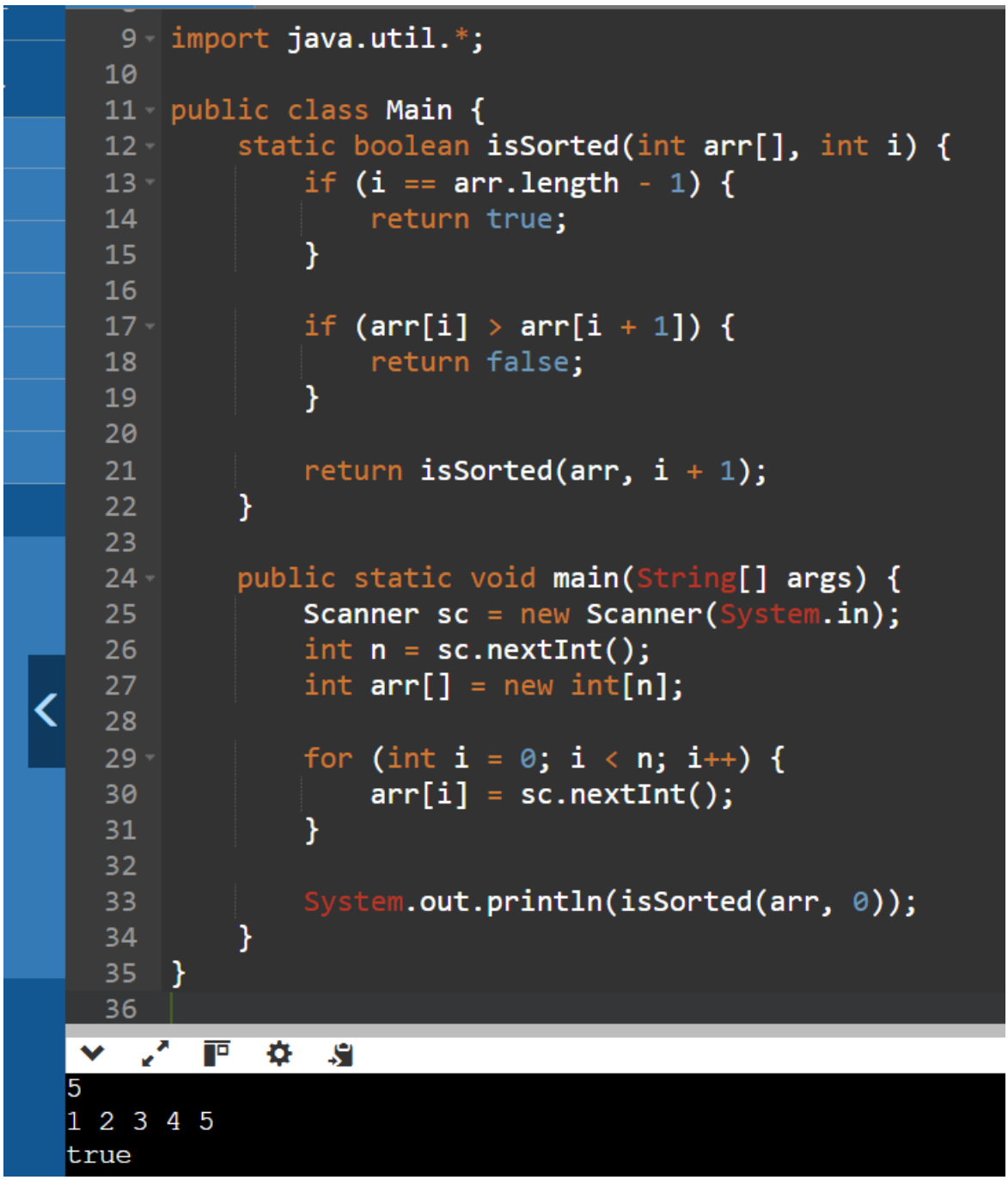
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        char oldChar = sc.next().charAt(0);
        char newChar = sc.next().charAt(0);

        System.out.println(replace(s, oldChar, newChar, 0));
    }
}
```

input

```
banana
a
o
bonono
```

22) Check Whether an Array is Sorted



```java
import java.util.*;

public class Main {
    static boolean isSorted(int arr[], int i) {
        if (i == arr.length - 1) {
            return true;
        }

        if (arr[i] > arr[i + 1]) {
            return false;
        }

        return isSorted(arr, i + 1);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        System.out.println(isSorted(arr, 0));
    }
}
```

```
5
1 2 3 4 5
true
```

23) Find the Max Element in an Array

```java

import java.util.*;
public class Main {
    static int max(int arr[], int i) {
        if (i == arr.length - 1) {
            return arr[i];
        }
        int m = max(arr, i + 1);

        if (arr[i] > m) {
            return arr[i];
        }

        return m;
    }

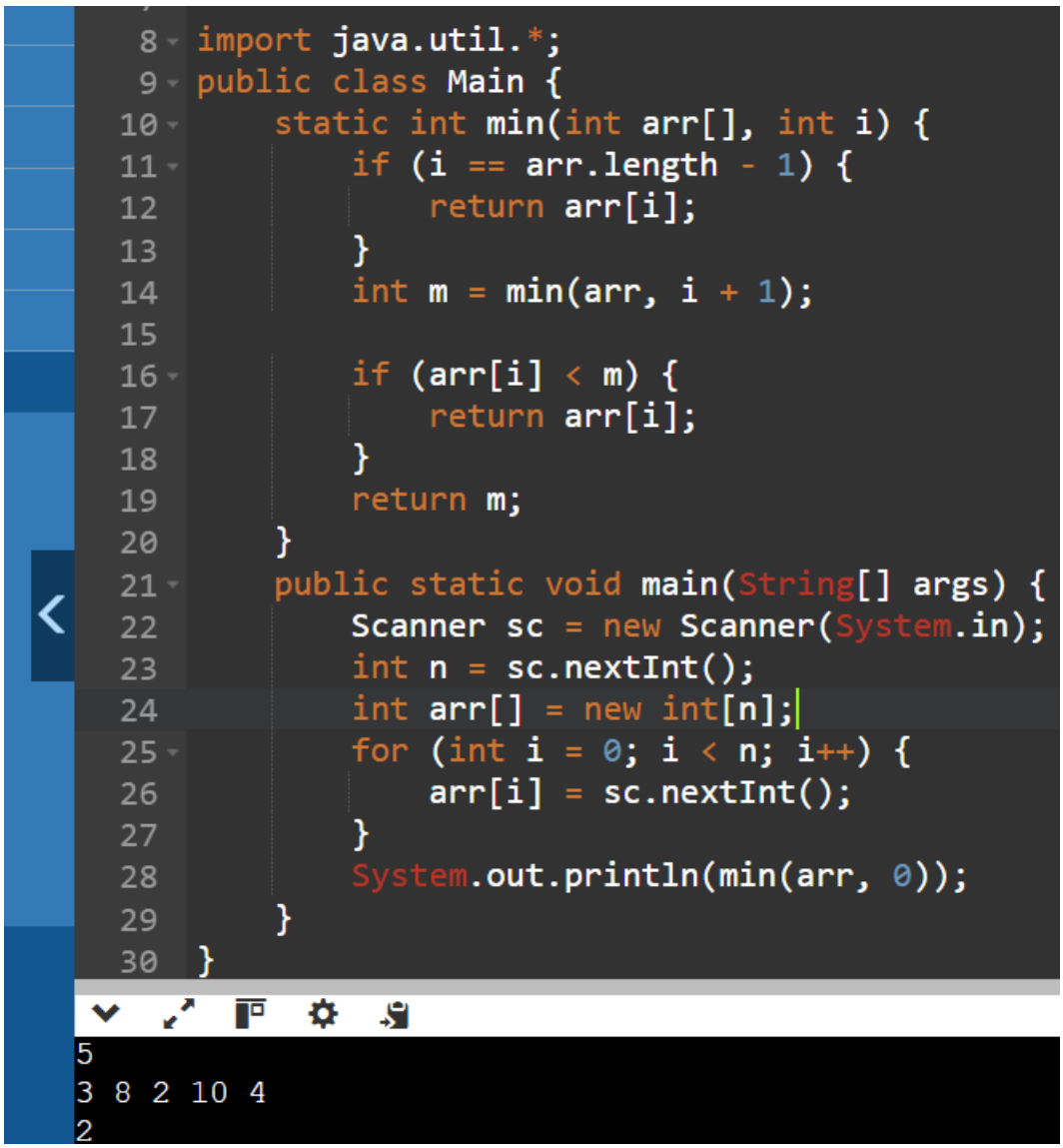
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        System.out.println(max(arr, 0));
    }
}
```

```
5
3 8 2 10 4
10
```

24) Find the Min Element in an Array



```
import java.util.*;
public class Main {
    static int min(int arr[], int i) {
        if (i == arr.length - 1) {
            return arr[i];
        }
        int m = min(arr, i + 1);

        if (arr[i] < m) {
            return arr[i];
        }
        return m;
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.println(min(arr, 0));
    }
}
```

```
5
3 8 2 10 4
2
```

25)Linear Search

```java
import java.util.*;
public class Main {
    static int search(int arr[], int i, int key) {
        if (i == arr.length) {
            return -1;
        }
        if (arr[i] == key) {
            return i;
        }
        return search(arr, i + 1, key);
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int key = sc.nextInt();
        System.out.println(search(arr, 0, key));
    }
}
```

```
5
10 20 30 40 50
30
2
```

26)Binary Search

```java
import java.util.*;
public class Main {
    static int search(int arr[], int low, int high, int key) {
        if (low > high) {
            return -1;
        }
        int mid = (low + high) / 2;
        if (arr[mid] == key) {
            return mid;
        }
        if (key < arr[mid]) {
            return search(arr, low, mid - 1, key);
        }
        return search(arr, mid + 1, high, key);
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int key = sc.nextInt();
        System.out.println(search(arr, 0, n - 1, key));
    }
}
```

input

```
5
10 20 30 40 50
40
3
```